

**Seat
No.**

Max. Marks: 30

- Instructions:** 1) All questions are compulsory.
2) Figures to the right indicate full marks.
3) Use of log table and calculator is allowed.

Q.1 Choose the Correct Alternative.

06

- The Arithmetic Mean of 87, 73, x , 48, 34 is 65.6 then value of x is _____.
 - 45
 - 86
 - 80
 - 84
- _____ cannot be determined graphically.
 - A.M.
 - Median
 - Mode
 - All of these
- Testing of human blood by taking few drops from body is an example of _____.
 - SRSWOR
 - SRSWR
 - Systematic sampling
 - Stratified sampling
- Coefficient of variation is always represented in _____.
 - negative
 - positive
 - percentage
 - None of these
- Which of the following average is also known as positional average?
 - A.M.
 - Median
 - Standard deviation
 - None of these
- If the profits of a company remain the same for the last ten months, then the Standard Deviation of profits for these ten months would be _____.
 - Positive
 - Negative
 - Zero
 - None of these

Q.2 Answer the following questions. (Any Three)

06

- Define term Population with examples.
- Find the median of the observation $X_i = 7, 3, 8, 7, 8, 6, 4, 8, 3, 4$
- Define positive correlation.
- Find range and coefficient of range of the observation $X_i = 22, 8, 19, 95$
- Write merits of Mode.

Q.3 Answer the following questions. (Any Two)**06**

- a) Write a note on Stratified Random Sampling.
 b) Find the value of mean of the following frequency distribution.

X	10	12	14	16	18
F	5	7	18	22	10

- c) Find correlation coefficient between X and Y.

X	10	14	21	22	30
Y	47	41	30	18	9

Q.4 Answer the following questions. (Any Two)**06**

- a) Find the value of median.

Marks	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50
No. of students	15	17	22	30	17

- a) Find the value of coefficient of variation of the
 $X_i = 73, 22, 87, 19, 86, 95$
 c) What is dispersion? Distinguish between absolute and relative measure of dispersion.

Q.5 Answer the following question. (Any One)**06**

- a) Draw the histogram from the following data. Hence determine the mode.

Classes	20 – 40	40 – 60	60 – 80	80 – 100	100 – 120	120 – 140
Frequency	19	25	32	38	48	11

- b) Write down the least square regression line procedure of Y on X.